Gawtam Chithra Ramesh

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Summary

Machine-learning researcher (MSc Systems, Control & Robotics, KTH) working on **reinforcement learning** and **end-to-end learning** for embodied agents, with reinforcement learning the through-line of my work since 2020. My experience spans **multi-agent PPO** policies for games and a **generative flow-matching** trajectory planner in my Master's thesis at ABB Robotics. Comfortable across **generative modeling**, deep and multi-agent reinforcement learning, and **motion planning and control**, with an independent, rapid-prototyping research style and three lead-author publications.

Education

KTH Royal Institute of Technology — Aug 2024 – Jul 2026
Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0) — *Stockholm, Sweden*

Indian Institute of Technology Madras — Aug 2019 – Jul 2024
Dual Degree (B.Tech in Engineering Design and M.Tech in Robotics) (GPA 7.9/10) — *Chennai, India*

Publications

- "Robust Object-layer Construction of 3D Scene Graphs Using Instance Segmentation." *International Conference on Image Processing and Vision Engineering (IMPROVE 2026).* (Accepted, Long Oral, Best Paper Award)
- "Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction." *ICPR 2026 Workshop on Perception, Interaction and Decision-Making for Human Cyber-Physical Systems (PIDM4HCPS)* . (Full paper, Accepted)
- "Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided Vision-Language Models." *IROS 2025 Workshop on Robotic Manipulation of Deformable Objects (ROMADO).* (Short paper, Accepted)

Professional Experience

Master's Thesis on Generative Planning for Long-Horizon Manipulation — Jan 2026 – Jun 2026 (Ongoing)
ABB Robotics, Mentors: Matthew Lock and Jonathan Styrud — *Västerås, Sweden*

- Building a **generative flow-matching trajectory planner** that learns a motion prior from offline demonstrations and is steered at **inference time** toward new objectives, with no retraining or per-task expert data.
- Conditioning one model on objectives at test time so a single planner transfers across task variants, rather than training a separate **imitation-learning** policy per task.
- Using **differentiable specification gradients** (Signal Temporal Logic) to steer generation toward compositional, temporally-extended goals.
- Validating on **long-horizon planning benchmarks** against state-of-the-art baselines, and transferring the approach to **6-DoF manipulation** with ABB Robotics R&D.

Device Research Intern on 3D Scene Graph — Jul 2025 – Dec 2025
Ericsson Research, Mentors: Puren Guler and Hector Caltenco — *Lund, Sweden*

- Developed a robust 3D scene-graph pipeline by integrating **instance segmentation and tracking** into a SOTA framework (Hydra), reducing sensitivity to parameter tuning across diverse environments.
- Designed a synchronized **RGB-D inpainting** module for privacy-aware 3D scene-graph construction, maintaining consistency across RGB and depth channels.
- Analyzed 3DSG frameworks to identify research gaps in object merging, parameter sensitivity, and privacy protection.
- Benchmarked SOTA inpainting algorithms to balance reconstruction fidelity with real-time performance.

Graduate Robotics Intern on Modular Collaborative Robots — Dec 2022 – Jul 2023
Systemantics India Pvt. Ltd., Mentor: Jagannath Raju — *Bangalore, India*

- Developed backward-chained **Behavior Trees** for a 6-DoF manipulator to enable robust, long-horizon trajectory planning.
- Designed and implemented a 6-DoF **trajectory generation and control** pipeline in C++, achieving **real-time** joint actuation via low-latency communication with motor controllers using D-Bus and SocketCAN.

Research Intern, Multi-Object Tracking — Jun 2021 – Jul 2021
Blurgs Research Labs Pvt. Ltd. — *Remote, India*

- Analyzed 5 state-of-the-art multi-object tracking networks for the startup's first product for the Indian Navy.
- Trained and tested multi-object tracking networks (**Siam-MOT**, **FairMOT**) for surveillance with drones and CCTV.
- Used **FairMOT** to track and trail multiple moving objects and re-identify them in real time.

Graduate Teaching Assistant Aug 2023 – Nov 2023
Field and Service Robotics (ID4060), Mentor: Bijo Sebastian *IIT Madras, India*

- Involved in design and evaluation of assignments based on ROS and CoppeliaSim for 40 students.
- Held meetings with small groups of students with diverse backgrounds to review their assignments and projects.

Strategist, Mentor & Co-organizer Apr 2020 – May 2021
Computer Vision and Intelligence Group (CVIG), IIT Madras *Chennai, India*

- Co-organized a deep-learning summer school for 200+ students spanning **deep learning**, **reinforcement learning**, and **computer vision**.
- Co-mentored student projects on **reinforcement learning** for games, an early thread of the RL focus that runs through my work today.

Research Experience

Graduate Robotics Research on Deformable Object Manipulation Jan 2025 – Nov 2025
Robotics Perception and Learning, Advisor: Florian T. Pokorny *KTH, Sweden*

- Integrated taxonomy-guided **vision-language models** (Gemini, GPT-4o, Qwen) to interpret deformable-object scenes and map their structured outputs to motion primitives.
- Built a prompt and taxonomy framework generating structured, robot-parsable specifications from natural-language instructions for downstream manipulation.
- Performed quantitative failure-mode analysis to guide iterative refinement of the taxonomy and prompts, integrating multimodal and temporal cues.

Dual Degree Project on Structure from Motion Jan 2024 – May 2024
INSPIRE Lab, Advisor: Nirav Patel *IIT Madras, India*

- Developed a **3D perception** system using COLMAP (**Structure from Motion**) from monocular RGB images to build cost-efficient human torso reconstruction for more accessible ultrasound.
- Developed an autonomous ultrasound scanning system using ROS, integrating a 5-DoF robotic arm with a single RGB camera to achieve cost-effective 3D perception for medical imaging.
- Implemented motion planning and control using ROS and MoveIt, and simulated the 5-DoF manipulation in Gazebo.
- Integrated perception algorithms within ROS, processing 3D point clouds from RGB images to enable accurate trajectory planning for the ultrasound probe.

Offline And Online Motion Planning & Control of Unmanned Aerial Vehicles Sep 2021 – Apr 2022
Young Research Fellowship, Advisor: *Prof. Satadal Ghosh* *IIT Madras, India*

- Developed a **target-driven motion planning** and **obstacle avoidance** algorithm for UAVs.
- Used Proportional Navigation and Acceleration-Velocity Obstacles in a 2D field to achieve a 20% faster and more compute-efficient algorithm than classical aerospace methods.

Selected Projects

Deep Learning, Advanced: Generative & Self-Supervised Models | DD2610, KTH Nov 2025 – Jan 2026

- Reimplemented **MeanFlow**, a one-step **generative flow-matching** model, from scratch in **PyTorch**; ablated time samplers, classifier-free guidance (ω, κ, p_c), and the $r \neq t$ ratio, reaching best FID ≈160 on CIFAR-10 (epoch 600) and 75% optimal $r{=}t$ mixing on MNIST – directly informing my thesis on generative trajectory planning.
- Built **Masked Autoencoders** with a **Vision Transformer** encoder from scratch (75% patch masking, MSE on masked patches), reducing reconstruction loss from 0.385 to 0.127 over 5 epochs on CIFAR-10.
- Implemented **contrastive** (**SimCLR**, NT-Xent τ=0.07) and **semi-supervised** (**FixMatch**, confidence threshold 0.95) learners in **JAX** with a ResNet-18 backbone, reaching ∼24% linear-probe accuracy from only 1% of labels.

Robot Learning and Embodied AI | DD2600, KTH Sep 2025 – Oct 2025

- Implemented a 3D semantic object mapping pipeline using RGB-D data, integrating **open-vocabulary object detection**, segmentation, and **CLIP** for semantic grounding.
- Deployed a pre-trained **Vision-Language-Action (VLA)** model for robot manipulation, executing **policy inference** from visual inputs and natural-language commands.
- Characterised generalisation and robustness failures of the VLA on unseen tasks, documenting the conditions under which it broke down.

Multi Agent Path Finding in Unity3D | DD2438, KTH — Feb 2025 – May 2025

- Developed a **Hybrid A*** path planner and waypoint tracking system in Unity3D for kinematically constrained single-agent (car/drone) navigation, achieving competitive performance.
- Engineered **Conflict-Based Search (CBS)** to solve multi-agent path finding for over 20 holonomic (drones) and non-holonomic (cars) agents, minimizing makespan in shared environments.
- Implemented a greedy dynamic goal assignment strategy for multi-agent teams (”bakery problem”) integrated with CBS, optimizing task allocation and minimizing overall travel time.

Reinforcement Learning for Pacman Capture The Flag | DD2438, KTH — Mar 2025 – May 2025

- Designed a hybrid AI for Pacman CTF (Unity3D), merging a custom **Behavior Tree** for high level strategy with **RL** models for specialized actions (attack, defend, escape).
- Trained **RL (PPO)** policies via **Unity ML-Agents** with role-specific designs (observations, rewards) and advanced methods (phased training, **self-play, curriculum learning**) for adaptive agent behavior.
- Evaluated strategic agent decision-making and adaptive behavior switching in a competitive, partially observable, dynamic **multi-agent** Pacman environment.

Image analysis and Computer vision | DD2423, KTH — Oct 2024 – Nov 2024

- Applied **Fourier Transforms** for image filtering and frequency analysis; designed Gaussian smoothing filters via FFT to reduce noise and study effects on varying image resolutions.
- Implemented a multi-scale differential geometry edge detector (computing L_{vv}, L_{vvv} for zerocrossings) and used Hough Transform for line detection from edges.
- Implemented image matching (**SIFT**, **RANSAC**) to estimate homographies (planar scenes) and fundamental matrices, enabling 3D scene reconstruction via triangulation

Introduction to robotics | DD2410, KTH — Sep 2024 – Oct 2024

- Implemented an iterative **inverse kinematics** solver from scratch for a 7-DoF KUKA robot arm in ROS.
- Applied **RRT*** to find collision-free paths for a **Dubins car**, showcasing path planning for non-holonomic vehicles.
- Engineered 2D **occupancy grid mapping** solutions in ROS, processing laser scans to build maps, identify free space, and generate C-spaces for navigation.
- Integrated a **Behavior Tree** mission planner for a TIAGo mobile manipulator (using ROS and Gazebo), enabling autonomous navigation, vision-based grasping, and robust task execution (e.g., kidnapped robot recovery).

Control of Automotive Systems | ED4060, IIT Madras — Jun 2022 – Nov 2022

- Designed a **heading-angle controller** for an **autonomous ground vehicle** from its **vehicle dynamics**, reducing error between the desired and actual heading angle within given performance specifications.
- Improved performance of a pneumatic brake system by designing **P and PI controllers** to meet given specifications.
- Designed a nonlinear **Sliding Mode Controller** to attenuate the effect of external forces on a tractor’s hydraulic hitch, reducing excessive vibrations and preventing front-wheel lift-off.

Multi-Agent Reinforcement Learning Shooter | CFI, IIT Madras — May 2020 – Aug 2020

- Trained cooperative **multi-agent PPO** policies in Unity3D for a team-based shooter, learning to outplay human opponents.
- Integrated the trained agents into a VR build; one of my first **reinforcement-learning** systems and the start of a long-running interest in learned multi-agent behaviour.

Technical Skills

- **Languages** : Python, C, C++, C#
- **Robotics** : ROS2, Unity3D, MATLAB, MuJoCo
- **DL** : PyTorch, JAX, OpenCV, wandb
- **Tools** : VS Code, Docker, Git, LaTeX, CUDA, Slurm

Relevant Coursework

Machine Learning: Robot Learning & Embodied AI, Deep Learning Advanced, Artificial Intelligence & Multi-Agent Systems, Fundamentals of Deep Learning
Robotics & Vision: Introduction to Robotics, Image Analysis & Computer Vision, Applied Estimation, Field and Service Robotics
Controls: Modelling of Dynamical Systems, Control Theory Advanced & Practice, Control of Automotive Systems

Achievements

- Awarded the IIT Madras Young Research Fellowship (top 30 of 800+) for outstanding research potential.
- JEE Advanced 97.64 percentile and JEE Mains 99.10 percentile (1.5M candidates); UCEED All-India Rank 52 of 50,000+.